# Solutions to Problem Set 2, Second Half

180.301 Microeconomic Theory - Spring 2026

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1. (a) $100 - 2y = 10 + 4y$, so quantity $y = 15$. price $P = 70$.

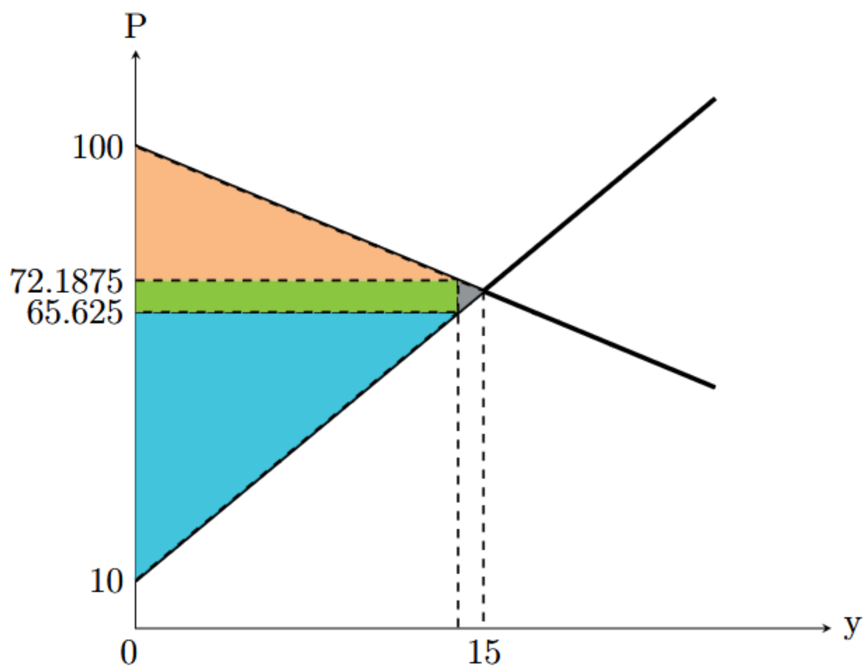
   (b) With an ad valorem tax of 10%, the price that the buyer pays is 10% higher than the price that the seller receives. So, $P_D = 1.1P_S \;\Rightarrow\; 100 - 2y = 1.1(10 + 4y)$ $\Rightarrow\; y = \frac{89}{6.4} = 13.90625,\; P_D = 100 - 2y = 72.1875,\; P_S = 10 + 4y = 65.625$.

   (c) Tax revenue is the green part of the graph below: $(P_D - P_S)y = 6.5625 \times 13.90625 \approx 91.26$.

   (d) Consumer surplus is the orange part of the graph: $CS = (100 - 72.1875) \times 13.90625/2 \approx 193.384$.

   Producer surplus is the blue part of the graph: $PS = (65.625 - 10) \times 13.90625/2 \approx 386.768$.

   (e) Deadweight loss is the gray part of the graph: $DWL = (15 - 13.90625) \times (72.1875 - 65.625)/2 \approx 3.589$.



2. (a) Let $P_D$ be the price paid by the buyer, and $P_S$ be the price received by the seller.

   Taxing sellers: the seller remits tax $0.01P_D$, so the seller keeps $P_S = 0.99P_D \Rightarrow P_D = \frac{1}{0.99}P_S \approx 1.0101P_S$.

   Taxing buyers: buyer must pay $P_D = 1.01P_S$.

   So, the equilibrium outcomes will be different. Taxing sellers is not equivalent to taxing buyers.

   (b) Taxing buyers is better for welfare, because the "tax wedge" of taxing buyers is $0.01P_S$, whereas the tax wedge of taxing sellers is $0.0101P_S$. The tax burden is a bit lower when we tax buyers.

   (c) Let $p$ be the "sticker price" of the good. The new tax on the sellers is $t'(p) = 0.01p+0.01$, so the net price that sellers actually receive is $P_S = p-(0.01p+0.01) = 0.99p - 0.01$. Buyers get a subsidy of $0.01p$, so the net price that buyers pay is $P_D = p - 0.01p = 0.99p$. We have $P_D - P_S = 0.01$. Compare this with the old tax on sellers: $P_D - P_S = 0.01p$. At the end of the day, the new government schemes just replaced the 1% proportional tax with a constant tax of \$0.01 per unit.

   So, this adjustment is effectively a tax cut if $p > 1$, and is effectively a tax rise if $p < 1$. In other words, the adjustment achieves its intended purpose of increasing equilibrium quantity if $p > 1$, and does not achieve the purpose if $p < 1$.